

Furthermore, the vertices of Δ^{in} are determined by the numbers d_i^j satisfying (*) and the following three equations

$$(**) \quad d_i^j - d_k^j = 0, i \neq k \neq j.$$

If three points $\bar{y}_i \in e_i$ have $|\bar{y}_i - \bar{y}_j| \leq \delta$, then, by the triangle inequality, the corresponding distances $\bar{d}_i^j$ satisfy

$$(***) \quad |\bar{d}_i^j - \bar{d}_k^j| \leq \delta \text{ for } i \neq k \neq j.$$

Since $\bar{d}_i^j$ also satisfy (*) the solution $\{d_i^j\}$ of (*) + (**) satisfies (by an easy computation)

$$|\bar{d}_i^j - d_i^j| \leq \delta/2.$$

Hence, by the triangle inequality

$$\text{insize } \Delta \leq 2\delta$$

and the lemma follows.

6.6.B Corollary: *If all geodesic triangles Δ in a geodesic metric space X have minsize $\Delta \leq \delta$, then X is 8δ -hyperbolic.*

6.6.C Let us relate minsize Δ to the area of the minimal disk D in X with $\partial D = \Delta$.

6.6.D Lemma: *Let $\Delta_0 \subset \mathbb{R}^2$ be a regular triangle with the edge length d_0 . If minsize $\Delta \geq \sqrt{3/2} d_0$ then there exists a short map $f: X \rightarrow \mathbb{R}^2$ such that*

(a) *The vertices x^j of Δ go to the vertices x_0^j of Δ_0 and the edges of Δ go to the corresponding edges of*

Δ_0 .

(b) The image of f lies inside Δ_0 .

Proof: Start with the map $F: X \rightarrow \mathbb{R}^3$ given by

$$F(x) = (1/\sqrt{3})(\text{dist}(x, e_1), \text{dist}(x, e_2), \text{dist}(x, e_3))$$

for the tree edges e_i of Δ . Clearly the map F is short, the image of F lies in the region

$$\{z_1, z_2, z_3 \mid z_i \geq 0, z_1 + z_2 + z_3 \geq \bar{\delta}/\sqrt{3}\} \subset \mathbb{R}^3 \text{ for } \bar{\delta} = \text{minsize } \Delta,$$

and the edge e_i goes to the plane $z_i = 0$ in $\mathbb{R}^3$ for $i = 1, 2, 3$. We compose F with the radial projection from the origin of $\mathbb{R}^3$ to the simplex

$$\bar{\Delta}_0 = \{z_1, z_2, z_3 \mid z_i \geq 0, z_1 + z_2 + z_3 = d_0/\sqrt{3}\}$$

thus obtaining the required map f .

6.6.E Corollary: Let D be a compact surface in X with $\partial D = \Delta$. Then

$$\text{area } D \geq (1/2\sqrt{3})(\text{minsize } \Delta)^2$$

Proof: Since Δ is homeomorphic to S^1 and $H_1(D, \Delta; \mathbb{Z}_2) \neq 0$, the map f sends D onto Δ_0 . Hence,

$$\text{area } D \geq \text{area } \Delta_0 = \sqrt{3}/4 d_0^2$$

(See [Gr1] and [Gr3] for sharper results of this kind.)

Remark: If a triangle Δ in X bounds a surface D , such that

$$\text{area } D \leq \epsilon_0^2 (\text{Diam } \Delta)^2,$$

then

$$(*) \quad \text{minsize } \Delta \leq \epsilon'_0 \text{ Diam } \Delta$$

for $\epsilon'_0 = \epsilon_0 \sqrt{2\sqrt{3}}$, as 6.6.E shows. One can prove by an elementary (and rather tedious) argument the following

6.6.F Proposition: *There exists a positive ϵ'_0 (one can take $\epsilon'_0 = 10^{-3}$) with the following property. If all geodesic triangles Δ in X with $\text{Diam } \Delta \geq R_0$ for a given $R_0 > 0$ satisfy $(*)$ then X is hyperbolic.*

We shall prove this (as well as Theorem 2.3.D) in Section 6.8 by using the Riemann mapping theorem and elementary properties of conformal metrics presented in the next section.

6.7 Analytic lemmas: Let $D(x,r) \subset \mathbb{R}^2$ be the disk $\{y \in \mathbb{R}^2 \mid \|x-y\| \leq r\}$ and $S(x,r)$ be the boundary circle $\partial D(x,r)$. Consider a positive function μ in the disk $D(x,R_0)$ for some $R_0 \geq 0$ and let

$$A(r) = A_\mu(x,r) = \int_{D(x,r)} \mu^2 dy$$

and

$$L(r) = L_\mu(x,r) = \int_{S(x,r)} \mu ds,$$

where $r \leq R_0$ and ds is the length parameter on the circle $S(x,r)$.

6.7.A *If*

$$(1) \quad L^2(r) \geq 4\pi c A(r)$$

for $r_0 \leq r \leq R_0$ and some r_0 and $c \geq 0$, then

$$(1') \quad A(r) \geq A(r_0)(r/r_0)^{2c}$$

for $r_0 \leq r \leq R_0$.

Proof: By Schwartz inequality the derivative A' of A satisfies

$$(2) \quad A'(r) = \int_{D(r)} \mu^2 \geq (2\pi r)^{-1} \left\{ \int_{S(r)} \mu \right\}^2 = (2\pi r)^{-1} L^2(r).$$

Then, by (1)

$$(3) \quad A'(r) \geq 2cr^{-1}A(r)$$

and (1') follows.

6.7.B Let $\Omega \subset \mathbb{R}^2$ be a bounded domain and set $d(x) = \text{dist}(x, \partial\Omega)$ for $x \in \Omega$. Next for a continuous positive function μ on Ω define $r(x, A)$ for $A \geq 0$ by the equation

$$A_\mu(x, r(x, A)) = A,$$

provided

$$A_\mu(x, d(x)) \geq A.$$

If $A_\mu(x, d(x)) < A$ we set $r(x, A) = d(x)$.

6.7.C Let some constants $a > 1$, $c > 1$ and $\alpha > 0$ satisfy

$$(4) \quad \frac{\frac{1}{a^c}}{a-1} = \alpha < 1.$$

Let $A_0 > 0$ be another constant, such that

$$(5) \quad \left(\frac{r(x, a^2 A_0)}{r(x, A_0)} \right)^{2c} \leq a^2,$$

for all those $x \in \Omega$, where $r(x, a^2 A_0) < d(x)$. Then

$$(6) \quad r(x, A_0) \geq (1-\alpha)a^{-\frac{1}{c}}d(x),$$

for all $x \in \Omega$.

Proof: Suppose (6) is violated by some $x_0 \in \Omega$ where

$$(6) \quad r_0 = r(x_0, A_0) < (1-\alpha)a^{-\frac{1}{c}}d(x_0).$$

To bring (6) to the contradiction we first observe that the average Av of $A(x, \bar{r}_1)$ over $x \in D(x, R_0)$ satisfies for $\bar{r}_1 = \alpha r_0$ and $R_0 = r(x_0, a^2 A_0)$.

$$(7) \quad Av \geq \frac{\bar{r}_1^2 A(x_0, R_0)}{(R_0 + \bar{r}_1)^2} \geq \frac{\bar{r}_1^2 a^2 A_0}{(r_0 a^{\frac{1}{c}} + \bar{r}_1)^2} = A_0,$$

as it follows from (5). Therefore, there exists a point $x_1 \in D(x_0, R_0) \subset \Omega$, such that

$$(8) \quad r_1 = r(x_1, A_0) \leq \bar{r}_1 = \alpha r_0.$$

Since

$$\|x_1 - x_0\| \leq R_0 \leq a^{\frac{1}{c}} r_0.$$

we obtain with (6),

$$(9) \quad d(x_1) \geq d(x_0) - \frac{1}{a^c} r_0 > \alpha d(x_0).$$

Hence x_1 also violates (6). Then we pass from x_1 to x_2, x_3 and so on and obtain a sequence $x_i \in \Omega$, such that

$$(10) \quad r_i = r(x_i, A_0) < \alpha^i r_0$$

and

$$\|x_i - x_{i+1}\| < \frac{1}{a^c} r_i \leq \frac{1}{a^c} \alpha^i r_0.$$

Hence, the points x_i lie in the disk $D(x_0, \rho)$ for some ρ , such that

$$\rho < \frac{1}{a^c} r_0 (1 + \alpha + \alpha^2 + \dots) = \frac{\frac{1}{a^c} r_0}{1 - \alpha} < d(x_0).$$

As the function μ is bounded on $D(x_0, \rho)$ the numbers r_i are bounded away from zero. But this contradicts to (10) as $\alpha < 1$ and the proof is concluded.

6.7.D Consider an equilateral triangle $\bar{\Delta}$ in $\mathbb{R}^2$ and a function $\bar{\mu}$ on $\mathbb{R}^2$ which is positive continuous on $\bar{\Delta}$ and zero outside $\bar{\Delta}$. Let the inequality

$$(11) \quad L_{\bar{\mu}}^2(x, r) \geq 4\pi c A_{\bar{\mu}}(x, r)$$

be satisfied by those $x \in \mathbb{R}^2$ and $r \geq 0$, for which

(a) the disk $D(x, r) \subset \mathbb{R}^2$ meets at most one edge of $\bar{\Delta}$;

$$(b) \quad A_0 \leq A_{\bar{\mu}}(x, r) \leq a^2 A_0,$$

where $c > 1$, $a > 1$ and $A_0 > 0$ are given constants.

Denote by $\bar{T} \subset \bar{\Delta}$ the union of the three segments in $\bar{\Delta}$ joining the center of $\bar{\Delta}$ with the centers of the edges of $\bar{\Delta}$. The length of these segments (that is the radius of the inscribed circle in $\bar{\Delta}$) is denoted by $\bar{d}$.

$$6.7.E \quad \text{Let } \alpha = \frac{\frac{1}{c}}{a-1} \text{ be } < 1. \text{ Then for every}$$

point $x \in \bar{T}$ there exists a number $\bar{r} = \bar{r}(x)$ in the interval

$$(12) \quad \bar{d} \geq \bar{r} \geq (1-\alpha)a^{-\frac{1}{c}}\bar{d},$$

such that the integral

$$A_{\bar{\mu}}(x, \bar{r}) = \int_{D(x, \bar{r})} \mu^2 dy$$

satisfies

$$(13) \quad A_{\bar{\mu}}(x, \bar{r}) \leq A_0.$$

Proof: Let $\bar{e}$ be the edge of $\bar{\Delta}$ nearest to a given point $x \in \bar{T}$. Consider the domain Ω which is the union of $\bar{\Delta}$ and the reflection of $\bar{\Delta}$ in $\bar{e}$. Now 6.7.E follows from 6.7.A and 6.7.C applied to an arbitrary small perturbation of $\bar{\mu}$ which makes $\bar{\mu}$ positive and continuous in Ω .

6.7.F Corollary: For every $x \in \bar{T}$ there exists a number r in the interval

$$(14) \quad \bar{d} \geq r \geq 1/2(1-\alpha)a^{-\frac{1}{c}}\bar{d},$$

such that the integral $L_{\bar{\mu}}(x,r)$ of μ over the circle $S(x,r)$ satisfies

$$(15) \quad L_{\bar{\mu}}(x,r) \leq 2\sqrt{\pi A_0}$$

Proof: Apply Schwartz inequality.

6.7.G Let $\bar{\Delta}_{\bar{\mu}}$ be the triangle $\bar{\Delta}$ with the Riemannian metric $\bar{\mu}^2(dx^2+dy^2)$.

6.7.G1 The minimal size of $\bar{\Delta}_{\bar{\mu}}$ satisfies

$$(16) \quad \text{minsize } \bar{\Delta}_{\bar{\mu}} \leq m_0 = 2(3+(1-\alpha)^{-1}a^{\frac{1}{c}})\sqrt{\pi A_0}.$$

Proof: For every piecewise smooth ark γ in $\bar{\Delta}$ between two points x_1 and x_2 in $\bar{\Delta}$ the $\bar{\mu}$ -distance $|x_1-x_2|_{\bar{\mu}}$ and the $\bar{\mu}$ -length $L_{\bar{\mu}}(\gamma) = \int_{\gamma} \bar{\mu}(x)d\gamma$ satisfy

$$|x_1-x_2|_{\bar{\mu}} \leq L_{\bar{\mu}}(\gamma).$$

Therefore, in order to prove 6.7.G1 one must find three arks γ_i between three points on the edges of $\bar{\Delta}$, such that

$$L_{\bar{\mu}}(\gamma_i) \leq m_0 \text{ for } i = 1,2,3.$$

This is done by first covering the tripod $\bar{T} \subset \bar{\Delta}$ by the minimal number of disks with centers in $\bar{T}$ whose radii r satisfy (14) and $\bar{\mu}$ -lengths of the boundary circles satisfy (15). The total number N of the disks obviously satisfy

$$N \leq N_0 = 3(1+(1-\alpha)^{-1}a^{\frac{1}{c}}).$$

Now, one constructs an inscribed triangle using arks of boundary

circles as in Figure 6 below

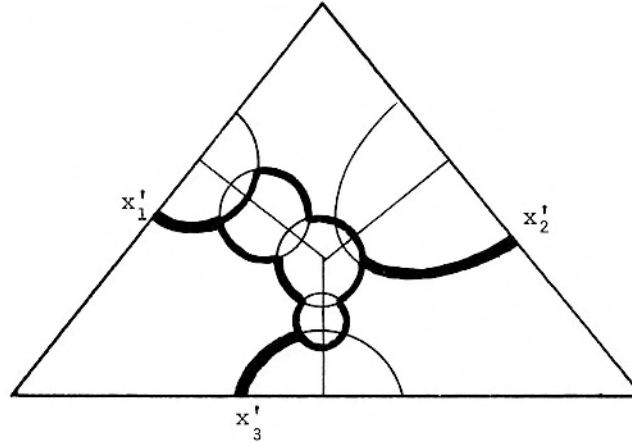


Figure 6

Notice that $\bar{\Delta}'$ degenerates at the circles adjacent to the edges of $\bar{\Delta}$, but in any case, this $\bar{\Delta}'$ obviously has the sides γ_i with

$$L_{\mu}(\gamma_i) \leq m'_0 = 2N_0\sqrt{\pi A_0}.$$

The number m'_0 is about three times greater than m_0 in (16). The (obvious) improvement of the argument giving us m_0 rather than m'_0 (which is not essentially important) is left to the reader.

6.8 Minimal surfaces, conformal maps and a length-area criterion for hyperbolicity: Consider a pair (V, f) where V is a smooth n -dimensional manifold and f is a continuous map of V into a metric space X . We call such a pair a (singular) *manifold* V in X and we define the *volume* of V by considering all those Riemannian metrics g on V for which the map $f: (V, g) \rightarrow X$ is short (i.e. distance non-increasing) and then by taking the infimum of total Riemannian volumes of the manifolds (V, g) , that is

$$\text{Vol}_X V = \inf_g \text{Vol}(V, g).$$

We are especially interested in the cases of *curves* in X that are

1-dimensional manifolds and of surfaces where $n = \dim V = 2$. The volume of a curve is, as usual, called the length and for surfaces it is the area.

Example: By projecting the unit circle $S^1 \subset \mathbb{R}^2$ to a coordinate axes one gets a circle S^1 in $\mathbb{R}$ of length = 2.

Next we define *the (filling) area* of a circle S^1 in X as the lower bound of areas of disks D in X bounded by S^1 (that are maps $D \rightarrow X$ extending the structure map $S^1 \rightarrow X$ from $S^1 = \partial D$).

6.8.A Let P be a polyhedron with a simplicial metric. A manifold V in P is called *simplicial* if the implied map $V \rightarrow P$ is simplicial for some triangulation of V . The volume of a simplicial V in X (obviously) equals the sum of volumes of n -simplices in P covered by the map $V \rightarrow P$, where each simplex is counted as many times as it covered by $V \rightarrow P$.

6.8.B In what follows P is a 2-dimensional polyhedron where all 1-simplices have the same length denoted d . Every simplicial curve S in P has length $L(S)$ which is an integer multiple of d , that is $L(S) = N_1 d$ and the area of every simplicial surface D in P is

$$A(D) = N_2 \frac{\sqrt{3}}{4} d^2 \text{ for some } N_2 = 0, 1, 2, \dots$$

Observe that every simple closed curve, that is a circle $S \subset P$, can be homotoped to a simplicial circle S' in P whose length $L(S')$ and area $A(S')$ are related to those of S by the inequalities

$$(17) \quad L(S') \leq 2L(S)$$

and

$$(17') \quad |A(S') - A(S)| \leq (1/2)dL(S).$$

This is done by deforming the intersections of S with the 2-simplices in P to the boundaries of these simplices as it is shown in Figure 7.

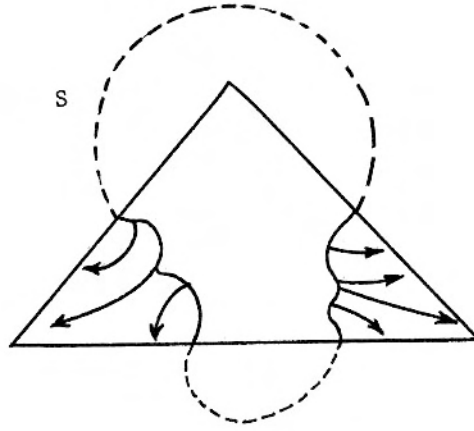


Figure 7

6.8.C Lemma: Let $A_0, A_1 \geq 4A_0$ and $C \leq A_0 d^{-2}$ be positive numbers, such that every simplicial circle S' in P with the area in the interval

$$2A_0 \leq A(S') \leq (1/2)A_1$$

satisfies

$$(18) \quad (L(S'))^2 \geq 8CA(S').$$

Then every (continuous) circle S in P with

$$A_0 \leq A(S) \leq A_1$$

satisfies

$$(19) \quad (L(S))^2 \geq CA(S).$$

Proof: Homotope S to S' which satisfies (17) and (17'). Since the inequalities $dL(S) \geq A(S)$ and $C \leq A_0 d^{-2} \leq d^{-2}A(S)$ imply (19), we may assume that $dL(S) \leq A(S)$. Then the inequality (17') implies

$$2A_0 \leq A(S') \leq 1/2A_1$$

and (19) follows from (17), (17') and (18). Q.E.D.

6.8.D Minimal surfaces: A disk D in X is called *minimal* if

$$\text{area}_X D = \text{area}_X \partial D < \infty.$$

If X is a "reasonable" space then every contractible circle S in X of finite area bounds a minimal disk. For example, if X is a two-dimensional polyhedron with a simplicial metric where all 1-simplices have the same length d , then, obviously, every simplicial curve S bounds a minimal disk D in X . In fact, one can find a *simplicial* minimal disk D with $\partial D = S$. On the other hand if the 1-simplices have different lengths, then this need not be true as simple examples (of 2-dimensional not locally finite polyhedra) show.

To avoid the existence problem we consider ϵ -minimal disks D , where $\text{area } D \leq \text{area } \partial D + \epsilon$ for a small positive ϵ which eventually goes to zero. This ϵ -minimality is good enough for our application given below.

6.8.E Conformal maps: A disk D in X is called *conformal* with respect to a fixed Riemannian metric g_0 on D if there exists a non-negative Borel function μ on D , such that the X -length of every smooth curve $S \subset D$ equals the μ -length of S that is the integral $\int_S \mu ds$ for the Riemannian length ds on S , and the

$$X\text{-area of every open } \Omega \subset D \text{ equals the } \mu\text{-area} = \int_{\Omega} \mu^2.$$

Consider a curvilinear triangle Δ in X that is a circle subdivided into three arcs or edges. Take a plane regular triangle $\bar{\Delta}$ (in place of D) and then look at *triangles $\bar{\Delta}$ in X bounded by Δ* . These by definition are continuous maps $f: \bar{\Delta} \rightarrow X$ sending the i -th edge of $\bar{\Delta}$ onto the i -th edge of Δ for some labeling of the edges by $i = 1, 2, 3$. In most "reasonable" cases the Riemann mapping theorem ensures the existence of a *conformal* minimal (or ϵ -minimal) $\bar{\Delta}$ in X bounded by a given contractible Δ . This is so,

for example, if Δ is a simple closed curve in the 1-skeleton of the polyhedron P in 2.6.A. In any case, Riemann's theorem provides, by a trivial approximation argument, an ϵ -minimal map $f: \bar{\Delta} \rightarrow X$ bounded by Δ , such that f also is ϵ -conformal, for some positive continuous function $\bar{\mu}$ on $\bar{\Delta}$. Namely,

$$|\text{length}_{\bar{\mu}} S - \text{length}_X S| \leq \epsilon$$

for all circle arcs $S \subset \bar{\Delta}$ and

$$|\text{area}_{\bar{\mu}} \Omega - \text{area}_X \Omega| \leq \epsilon$$

for all open $\Omega \subset \bar{\Delta}$, where one can take $\epsilon > 0$ arbitrarily small. This, for $\epsilon \rightarrow 0$, allows us to use the analytic lemmas in 6.7 (applied to the extension of μ by $\equiv 0$ on $\mathbb{R}^2 \setminus \bar{\Delta}$) in so far as we can control the X -lengths of curves and areas in $\bar{\Delta}$. The pertinent curves are topological circles S in $\bar{\Delta}$ formed by two segments S_1 and S_2 , where S_1 lies in an edge of $\bar{\Delta}$ and S_2 is a circle arc in Δ (see Figure 8).

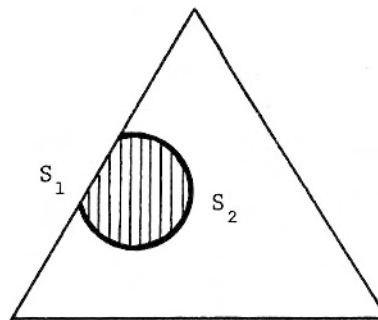


Figure 8

If, for example, the triangle Δ is geodesic, then

$$\text{length}_X S_1 \leq \text{length}_X S_2$$

and thus

$$(20) \quad \text{length}_X S \leq 2 \text{length}_X S_2.$$

This ensures a bound on the X -area of the disk $D \subset \bar{\Delta}$ bounded by S in term of $\text{length}_X S_2$, provided a similar bound for $\text{length } S$ is available. More generally, call Δ a λ -geodesic triangle for some $\lambda \geq 1$ if every two points x_1 and x_2 lying in an edge of Δ bound a segment γ in this edge, such that

$$\text{length}_\gamma \leq \lambda |x_1 - x_2|.$$

If Δ is λ -geodesic then instead of (20) we have

$$(20) \quad \text{length}_X S \leq (1+\lambda) \text{length}_X S_2,$$

which serves the same purpose as (20).

Example: Let every edge of Δ be a minimal geodesic in the 1-skeleton of the above polyhedron P . Then, obviously, Δ is 2-geodesic.

6.8.F The above discussion allows us to apply the results in 6.7, in particular, Lemma 6.7.6, to an arbitrary metric space X as follows. Let every circle S in X with the area in the interval

$$A_0 \leq A(S) \leq A_1,$$

for given positive numbers A_0 and $A_1 \geq A_0$ satisfy

$$L^2(S) \geq CA(S),$$

where $L(S)$ denotes the length of S . Let a number $\lambda \geq 1$ be given and suppose there exist numbers $a > 1$, $c > 1$ and $\alpha < 1$ with the following three properties

$$(i) \quad A_1 \geq a^2 A_0,$$

$$(ii) \quad C \geq 4c\pi(1+\lambda)^2,$$

$$(iii) \quad \alpha = \frac{\frac{1}{c}}{a-1}.$$

Now, Lemma 6.7.G₁ leads to the following

6.8.G Proposition: *Let Δ be a λ -geodesic triangle in X of finite area. Then the minimal size of Δ is bounded by*

$$\text{minsize } \Delta \leq m_0 = 2(3+a^{\frac{1}{c}}(1-\alpha)^{-1})\sqrt{\pi A_0}.$$

6.8.H Corollary: *If $A_1 \geq 16A_0$ and $C \geq 48(1+\lambda)^2$, then*

$$\text{minsize } \Delta \leq 24\sqrt{A_0}.$$

Proof: Take $a = 4$ and $c = \frac{C}{4\pi(1+\lambda)^2}$.

6.8.I Let us return to the 2-dimensional polyhedron P with the simplices of size d . Let every simplicial circle S' in P with the area in the interval

$$A'_0 \leq A(S') \leq 64A'_0$$

for some $A'_0 > 0$ satisfies

$$(L(S'))^2 \geq C'A(S')$$

for $C' = 4000$. Then consider a curvilinear triangle Δ in P , where each edge is either a geodesic segment in P or in the 1-skeleton P^1 of P . Notice that such a Δ is λ -geodesic in P for $\lambda = 2$.

By applying the above we obtain the following

6.8.J Theorem: *If Δ is contractible and $A'_0 \geq 500d^2$, then*

$$\text{minsize } \Delta \leq 24\sqrt{A'_0}.$$

This yields with 6.6,

6.8.K Corollary: *If P is simply connected (and, hence, every Δ is contractible) then P is δ -hyperbolic for $\delta = 200\sqrt{A'_0}$ and P^1 is 2δ -hyperbolic for the simplicial metric in P^1 with the simplices of size d .*

6.8.L Remark: Suppose the above polyhedron P equals the 2-skeleton of $P_d(X)$ for some geodesic metric space X (see 1.7). Then every geodesic triangle Δ in X gives rise to a geodesic triangle Δ' in the 1-skeleton $P^1 = P_d^1(X)$, such that

$$\text{minsize } \Delta' \geq \text{minsize } \Delta.$$

this leads to the following

6.8.M Length-area criterion for hyperbolicity: *Let for some $d \geq 0$ and $A'_0 \geq 500d^2$ every simplicial circle S' in $P_d^1(X)$ with the area in the interval*

$$A'_0 \leq A(S') \leq 64A'_0$$

satisfies

$$L^2(S') \geq 4000A(S),$$

where the areas are measured in $P_d^2(X) \supset P_d^1(X)$. If $P_d(X)$ is simply connected, then X is δ -hyperbolic for

$$\delta = 400/\sqrt{A_0}.$$

Observe that we have already proven (see 2.3) the converse which can be expressed in the present language as follows.

6.8.N *If X is δ -hyperbolic and $d \geq 8\delta$ then every $S' \subset P_d^1(X)$ satisfies*

$$L(S') \geq (d/4\sqrt{3})A(S').$$

6.8.P The proof of 2.3.D: If we apply the above results to the 2-dimensional polyhedron associated to the (triangular!) presentation of Γ and take into account the discrepancy factor $\frac{9^4}{\sqrt{3}}$ between the two areas, we immediately obtain 2.3.D with exception of the inequality $A(f) \leq CL(f)$. This as well as the (now obvious) proof of 2.3.F is left to the reader.

6.8.Q The proof of 6.6.F: The inequality (*) implies the δ -hyperbolicity of every ball B in X of radius $R \geq (\epsilon_0')^{-1}R_0$ for $\delta \leq 8\epsilon_0'R$. This implies an isoperimetric inequality for disks in this ball, which, in turn, yields the hyperbolicity of X .

6.8.R Let us characterize the hyperbolicity by the properties of conformal minimal maps. Assume X is either a Riemannian manifold or a locally compact finite dimensional polyhedron with a simplicial metric and let the isometry group $Is X$ be cocompact on X . A simple compactness argument shows that

6.8.S *The universal covering of X is hyperbolic if and only if every conformal minimal map $\mathbb{R}^2 \rightarrow X$ is constant (where a map $\mathbb{R}^2 \rightarrow X$ is called minimal if its restriction to every disk $D \subset \mathbb{R}^2$ is minimal).*

6.8.T This proposition suggests the following stronger form of hyperbolicity. Call a map f of a surface S into X *locally minimal* if each point $s_0 \in S$ is contained in a disk $D_0 \subset S$ such

that the restriction of f to D_0 is minimal. Notice that for 2-polyhedra X the local minimality is a (local) *topological* invariant of f .

Let X be a space as above. Then a simple argument yields

6.8.U *The following three conditions are equivalent.*

(1) *Every conformal locally minimal map $\mathbb{R}^2 \rightarrow X$ is constant.*

(2) *There exists a constant $C \geq 0$ such that every locally minimal disk D in X satisfies*

$$\text{Area } D \leq C \text{ length } \partial D.$$

(3) *There exists a constant $\lambda > 0$ such that every conformal locally minimal map $f: D(R) \rightarrow X$ for $D(R) = \{x \in \mathbb{R}^2 \mid \|x\| \leq R\}$ satisfies*

$$\text{Diam}_X f(D(1/2R)) \leq \lambda.$$

Remark: A space X satisfying (2) is called *locally hyperbolic* (which is stronger than hyperbolic). Notice that locally hyperbolic 2-polyhedra has been extensively studied (under various names) in the *small cancellation theory*. In particular, compact 2-dimensional polyhedra with $K < 0$ as well as hyperbolic small cancellation polyhedra in 4.7 are locally hyperbolic.

§7. Geodesics, quasigeodesics and quasiconvexity

Let us return to the study of δ -thin triangles in a geodesic metric space X (compare 6.3). Observe, that if $\Delta = [x_1, x_2, x_3]$ is a δ -thin geodesic triangle, then every edge of Δ , say $[x_1, x_3]$ is δ -close to the union of the remaining two edges. That is

$$(*) \quad \text{dist}(x, Y) \leq \delta$$

for $Y = [x_1, x_2] \cup [x_2, x_3]$ and all $x \in [x_1, x_3]$.

Now, let Y be a broken geodesic consisting of $n \leq 2^k$ edges, say $Y = [x_1, x_2] \cup [x_2, x_3] \cup \dots \cup [x_n, x_{n+1}]$. Then by applying (*) k times we obtain the following

Lemma: *If every geodesic triangle with the vertices in the subset $\{x_1, \dots, x_{n+1}\} \subset X$ is δ -thin, then every point x in X between x_1 and x_{n+1} (i.e. x lies in some geodesic segment $[x_1, x_{n+1}]$) satisfies*

$$\text{dist}(x, Y) \leq k\delta.$$

7.1 Exponential growth of balls: Take a ball $B \subset X$ around $x_0 \in X$ of radius d ,

$$B = \{x \in X \mid |x_0 - x| \leq d\}$$

and let a geodesic segment $[y, z] \subset X$ contain x_0 and have the end points y and z outside B . Let $f: [a, b] \rightarrow X$ for $a, b \in \mathbb{R}$ be a continuous path joining y and z in the complement $X \setminus B$. That is

$$f(a) = y, f(b) = z,$$

and

$$|f(t) - x_0| \geq d \text{ for all } t \in [a, b].$$

7.1.A If all triangles with the vertices in the set $f([a,b]) \subset X$ are δ -thin then the length of the path is bounded from below by

$$\text{length } f \geq \delta(2^{d/\delta}-2).$$

Proof: For every $n = 1, 2, \dots$ there exist, by the very definition of length, $n+1$ points in $[a,b]$, say $a = t_0 \leq t_1 \leq \dots \leq t_n = b$, such that

$$\delta_i \stackrel{\text{def}}{=} |f(t_i) - f(t_{i+1})| \leq L/n,$$

for $L = \text{length } f$ and $i = 0, 1, \dots, n-1$. Now we make $\delta_i \leq 2\delta$ for some $n = 2^k$, where $n - 1 \leq L/2\delta$, and apply 2.1.E" to some broken geodesic V with the vertices $y = f(t_0), f(t_1), \dots, f(t_n) = z$. This yields

$$d \leq \delta + \text{dist}(x_0, Y) \leq (k+1)\delta,$$

and

$$L \geq 2n\delta - 2\delta = (2^{k+1}-2)\delta \geq (2^{d/\delta}-2)\delta.$$

7.1.B Corollary: Consider a segment $[x,y] \subset X$ and a path $f: [a,b] \rightarrow X$ of finite length between x and y , such that

$$\text{dist}(f(t_0), [x,y]) \geq d_0$$

for some $d_0 \geq 20\delta$ and $t_0 \in [a,b]$. If all triangles with the vertices in the union $[x,y] \cup f[a,b]$ are δ -thin, then for every $\ell_0 \leq d_0$ there exists a subinterval $[a_0, b_0] \subset [a,b]$, such that the length L_0 of the path $f|_{[a_0, b_0]}: [a_0, b_0] \rightarrow X$ satisfies

$$(*) \quad \ell_0 \leq L_0 \leq 2\ell_0$$

and

$$(**) \quad L_0 \geq \delta(2^{\mu_0/\delta} - 2) \text{ for } \mu_0 = 1/2 |f(a) - f(b_0)| - 4\delta$$

Proof: Since the function $t \mapsto \text{dist}(f(t), [a, b])$ is continuous, there exists a subinterval $[a_0, b_0]$ satisfying (*) and such that

$$d_1 \stackrel{\text{def}}{=} \min \{ \text{dist}(f(a_0), [x, y]), \text{dist}(f(b_0), [x, y]) \} \geq d_0/2, \\ \text{dist}(f(t), [x, y]) \geq d_1 \text{ for all } t \in [a_0, b_0].$$

Take some segment $[x_0, y_0]$ for $x_0 = f(a_0)$ and $y_0 = f(b_0)$, and show that the center z_0 of $[x_0, y_0]$ satisfies

$$\text{dist}(z_0, f[a_0, b_0]) \geq \mu = \min(1/2 |x_0 - y_0| - 4\delta, 1/2 d_0 - 4\delta).$$

To see this take some (nearest) points x_1 and y_1 in $[x, y]$, such that

$$|x_1 - x_0| = \text{dist}(x_0, [x, y]) \text{ and } |y_1 - y_0| = \text{dist}(y_0, [x, y])$$

and use the fact that z_0 is 2δ -close to the broken geodesic $[x_0, x_1, y_1, y_0]$. Then

$$L_0 \geq (2^{\mu/\delta} - 2)\delta.$$

Since $L_0 \leq 2d_0$ and $d_0 \geq 20\delta$ we have

$$L_0 \leq (2^{\mu_1/\delta} - 2)\delta \text{ for } \mu_1 = 1/2 d_0 - 4\delta,$$

and (**) follows.

7.1.C A discrete version of 7.1.B: Consider a sequence of points $x_1, x_2, \dots, x_n$ in a δ -hyperbolic space X , such that $x_1 = x$ and $x_n = y$ for a given segment $[x, y]$ in X . Let $d_0 = \max_i \text{dist}(x_i, [x, y])$ and let $\text{dist}(x_{i_0}, [x, y]) \geq d_0 - \epsilon$ for some $i_0 = 2, 3, \dots, n-1$ and some

$\epsilon > 0$. Let

$$\mu_0 = \max(|x_{i_0-1} - x_{i_0}|, |x_{i_0} - x_{i_0+1}|).$$

If X is a tree (i.e. $\delta = 0$), and $\mu_0 \leq d_0/2 - \epsilon$, then, obviously,

$$|x_{i_0-1} - x_{i_0+1}| \leq \mu_0 + 4\epsilon.$$

Now, let $\delta > 0$. Then the subset $[x, y] \cup \{x_{i_0-1}, x_{i_0}, x_{i_0+1}\}$ in X can be 8δ -approximated by a tree (see 6.1) which leads to the inequality

$$|x_{i_0-1} - x_{i_0+1}| \leq \mu_0 + 4\epsilon + 40\delta.$$

Notice, that only the case $\epsilon = 0$ is needed for further applications but $\epsilon > 0$ is necessary when we pass to the approximating tree. In fact, we shall need later the following more technical fact which is obtained by a similar approximation argument.

7.1.D Lemma: Let $2 \leq i_0 \leq n-2$ and

$$\text{dist}(x_{i_0}, [x, y]) = d_0 = \max_i \text{dist}(x_i, [x, y]).$$

If

$$\max(|x_{i_0-2} - x_{i_0-1}| + |x_{i_0-1} - x_{i_0}|, |x_{i_0} - x_{i_0+1}| + |x_{i_0+1} - x_{i_0+2}|) \leq d_0/2$$

then one of the following three inequalities holds

- (1) $|x_{i_0-1} - x_{i_0+1}| \leq \max(|x_{i_0-1} - x_{i_0}|, |x_{i_0} - x_{i_0+1}|),$
- (2) $|x_{i_0-2} - x_{i_0+1}| - 100\delta \leq \max(|x_{i_0-2} - x_{i_0-1}|, |x_{i_0-1} - x_{i_0}|, |x_{i_0} - x_{i_0+1}|),$
- (3) $|x_{i_0-1} - x_{i_0+2}| - 100\delta \leq \max(|x_{i_0-1} - x_{i_0}|, |x_{i_0} - x_{i_0+1}|, |x_{i_0+1} - x_{i_0+2}|).$

7.2 Stability of quasigeodesics: A path $f: [a, b] \rightarrow X$ is

called λ -geodesic (or quasigeodesic if the specific value of the constant λ is irrelevant) if

$$\text{length } f([a',b']) \leq \lambda |f(a')-f(b')|,$$

for all $[a',b'] \subset [a,b]$. One immediately obtains with the previous discussion.

7.2.A Proposition: Every λ -geodesic path between two points x and y in a δ -hyperbolic geodesic metric space X stays ϵ -close to every geodesic segment $[x,y] \subset X$ for $\epsilon \leq C\delta(1+\log \lambda)$, where $C \leq 100$ is some universal constant.

7.2.B Remark: The above obviously generalizes to locally λ -geodesic paths f which are, by definition, λ -geodesic on all segments $[a',b'] \subset [a,b]$ with

$$\text{length } f([a',b']) \leq \text{const} = \text{const}(\delta, \lambda) \leq 1000\delta\lambda.$$

It follows that every local λ -geodesic is, in fact, a (global!) 2λ -geodesic.

Let us state a similar result for "discrete quasigeodesics" which easily follows from the discussion in 7.1.

7.2.C Let $x_1, x_2, \dots, x_n$ be some points in a δ -hyperbolic space X (which is not assumed geodesic any more), such that

$$|x_i - x_{i-2}| - \epsilon \geq \max(|x_i - x_{i+1}|, |x_{i+1} - x_{i+2}|)$$

for some $\epsilon > 100\delta$ and $i = 1, 2, \dots, n-1$. Then

$$\lambda |x_1 - x_n| \geq \sum_{i=1}^{n-1} |x_i - x_{i+1}|,$$

for

$$\lambda = (0.01\epsilon - \delta)^{-1} \max_i |x_i - x_{i+1}|.$$

7.2.D Let us show that the implication local $\Rightarrow$ global for quasigeodesics characterizes hyperbolic spaces. In fact we have the following stronger

7.2.E **Proposition:** Let X be a non-hyperbolic geodesic metric space and let $\epsilon_0 > 0$, $L_0 > \epsilon_0$, $L_1 > L_0$ and $\lambda > 1$ be given numbers. Then there exists a circle S^1 in X of finite length $L \geq L_1$, such that every ark $[a', b']$ in S^1 of length L' in the interval $\epsilon_0 \leq L' \leq L_0$ satisfies

$$(*) \quad |a' - b'|_X \geq \lambda^{-1} L'$$

Proof: If every circle S^1 in X contains an interval which violates $(*)$ then one can shorten S^1 by joining the ends of this interval by a geodesic segment in X . By repeating this process one "subdivides" S^1 into $k \leq \text{const}$ length circles of length $\leq L_0$, which yields the isoperimetric inequality in $P_d^2(X)$ for $d = L_0$ which implies hyperbolicity. Q.E.D.

7.2.F **Remarks:** One can easily sharpen 7.2.E by requiring S^1 to be an embedded circle in X . Furthermore, if X is complete one can allow $\epsilon_0 = 0$.

7.2.G **Quasiisometric maps:** A map between metric spaces, say $f: X \rightarrow Y$, is called

(a) An ϵ + isometric map for some $\epsilon \geq 0$ if

$$|x_1 - x_2| - \epsilon \leq |f(x_1) - f(x_2)| \leq |x_1 - x_2| + \epsilon$$

for all x_1 and x_2 in X . Notice that ϵ + isometry does not imply the continuity of f .

(b) An *isometric* $\times \lambda$ map for some $\lambda \geq 1$ if

$$\lambda^{-1} |x_1 - x_2| \leq |f(x_1) + f(x_2)| \leq \lambda |x_1 - x_2|.$$

(c) An ϵ + *isometric* $\times \lambda$ map if

$$\lambda^{-1} |x_1 - x_2| - \epsilon \leq |f(x_1) + f(x_2)| \leq \lambda |x_1 - x_2| + \epsilon.$$

Sometimes we suppress ϵ and λ and call such maps + *isometric* $\times$ or *quasiisometric*. In a similar way we speak of + *isometric* and *isometric* $\times$ maps.

Examples:

(a) If X is a geodesic space, then the canonical map $X \rightarrow P_d^1(X)$ is + *isometric* for every $d \geq 0$, where the polyhedron $P_d^1(X)$ is given the simplicial metric with the edges of length d . In fact the map $X \rightarrow P_d^1(X)$ is d + *isometric*.

(b) If $X \rightarrow Y$ is an *injective* + *isometric* $\times$ map then the induced map $P_d^1(X) \rightarrow P_d^1(Y)$ is *isometric* $\times$ for every $d \geq 0$. It follows that every + *isometric* $\times$ map between geodesic spaces, $f: X \rightarrow Y$, admits a decomposition

$$\begin{array}{ccccccc} & & & & f & & \\ & & & & \curvearrowright & & \\ X & \xrightarrow{f_1} & X_1 & \xrightarrow{\bar{f}_2} & Y_1 & \xrightarrow{f_2} & Y \end{array}$$

where f_1 is an *injective* + *isometric* map, $\bar{f}$ is *isometric* $\times$ and f_2 is *surjective* + *isometric*.

7.2.H Theorem: Let X and Y be hyperbolic spaces where X is geodesic. Then every + *isometric* $\times$ map $f: X \rightarrow Y$ (canonically) induces a topological embedding between the hyperbolic boundaries, say

$\partial f: \partial X \rightarrow \partial Y$. Furthermore, if the image $f(X) \subset Y$ is cobounded (i.e. $\text{dist}(y, f(X)) \leq \text{const} < \infty$ for all $y \in Y$) then ∂f is a homeomorphism.

Proof: Observe that the theorem is obvious for λ -isometric maps between arbitrary (i.e. non-geodesic) metric spaces. Then using 6.4 and the above discussion we reduce the problem to isometric λ -maps $\tilde{f}$ between hyperbolic geodesic spaces. Such maps $\tilde{f}$ (obviously) send geodesic to quasigeodesics and these lie within finite (and uniformly bounded from above) distance from geodesics. This allows us to control the behaviour of the "product" (x, y) which is equal, up to a bounded error, to the distance from the reference point to some geodesic segment between x and y . Now it is clear that "convergent" sequences $x_i \rightarrow a \in \partial X$ sent by f to "convergent" sequences $y_i \rightarrow b \in \partial Y$ and the proof follows from the definition of the hyperbolic boundary (see 1.8).

7.2.J The conclusion of 7.2.H holds true for *locally* $\epsilon + \lambda$ -isometric λ -maps f . These are, by definition, $\epsilon + \lambda$ -isometric λ on every ball B in X of radius $R \leq C(\delta + \epsilon)\lambda$ where δ is the implied hyperbolicity constant (serving X as well as Y) and $C \leq 10,000$ is some universal constant. This "localization" of 7.2.H follows from 7.2.B, which also shows that locally λ -isometric λ -maps are, in fact, (globally) λ -isometric λ .

7.2.K A conformal view on the boundary ∂X (compare 1.8 and [F1]). Let μ be a positive function on a geodesic metric space (X, d) . Define the μ -length of curves S in X as $\int_S \mu$. Next define the metric d_μ on X as the infimum

of the μ -lengths of paths between given points in X . Let $\bar{X}$ denote the metric completion of X and $\bar{X}_\mu$ be the completion of (X, d_μ) . Define the μ -boundary of X by $\partial_\mu X = \bar{X}_\mu \setminus \bar{X}$.

To relate $\partial_\mu X$ and ∂X (for some special μ) we need the following property of d_μ -geodesics and quasigeodesics which immediately follows from the local characterization of quasigeodesics

(see 7.2.B).

7.2.L Lemma: Let X be δ -hyperbolic and let

$$(*) \quad |\log \mu(x_1) - \log \mu(x_2)| \leq \epsilon_0$$

for all x_1 and x_2 in X satisfying $|x_1 - x_2| \leq \delta$ and for some (universal constant) $\epsilon_0 > 0.001$. Then every $||_\mu$ -quasigeodesic also is $||$ -quasigeodesic. Furthermore $||$ - and $||_\mu$ -geodesic segments between some points in X are $\epsilon_0^{-1}\delta$ -close for the original metric in X and the distance $|x-y|_\mu$ is related to the μ -length of each $||$ -geodesic segment between x and y by

$$\mu\text{-length} \leq \epsilon_0^{-1} |x-y|_\mu$$

for all x and y in X .

From this we immediately derive the following

7.2.M Corollary: Let

$$\mu(x) = \mu_{x_0}(x) = \exp - \epsilon |x - x_0|$$

for some ϵ in the interval $0 < \epsilon < 1/2 \epsilon_0 \delta^{-1}$. Then the identity map $(X, ||) \leftrightarrow (X, ||_\mu)$ continuously extends to a homeomorphism $X \cup \partial X \leftrightarrow X \cup \partial_\mu X$. Furthermore, if some points a and b in $\partial X = \partial_\mu X$ are limits of sequences x_i and y_i respectively for which the sequence (x_i, y_i) converges to some limit denoted (a, b) , then the $||_\mu$ -distance between a and b satisfies

$$|\epsilon(a, b) + \log |a-b|_\mu| \leq \epsilon_0^{-1} + \delta.$$

It follows that every isometry of X induces a $||_\mu$ -Lipschitz map on $(\partial X, ||_\mu)$ and every quasi-isometry is Hölder on ∂X .

7.3 Quasiconvex subsets: A subset Y in a geodesic space X is called ϵ -convex (or quasiconvex) for some $\epsilon \geq 0$ if every

geodesic segment $[y_1, y_2]$ with the end points in Y lies ϵ -close to Y . In other words, $[y_1, y_2]$ is contained in the ϵ -neighborhood of Y that is

$$Y + \epsilon = \{x \in X \mid \text{dist}(x, Y) \leq \epsilon\}.$$

Example: If X is δ -hyperbolic then every ball $B \subset X$ around some $x_0 \in X$ is 2δ -convex since $[y_1, y_2]$ is 2δ -close to the union of $[x_0, y_1]$ and $[x_0, y_2]$. More generally, we have

7.3.A Lemma: *If X is δ -hyperbolic and $Y \subset X$ is ϵ -convex then the ρ -neighborhood $Y + \rho$ is 4δ -convex for all $\rho \geq \epsilon$.*

Proof: Recall that every segment $[x_1, x_2] \subset X$ stays 4δ -close to the broken geodesic $[x_1, y_1, y_2, x_2]$, where y_1 and y_2 are *normal projections* (see 7.3.B below) of x_1 and x_2 to Y respectively. If x_1 and x_2 lie in $Y + \rho$ for $\rho \geq \epsilon$ then $[x_1, y_1, y_2, x_2]$ is contained in $Y + \rho$ and the proof follows.

7.3.B Definition: If for every $x \in X$ the function $y \mapsto |x - y|$ assumes the minimum on Y we assign to each x such a minimum point and thus define a *normal projection* (or *retraction*) called $P_Y: X \rightarrow Y$. In other words $P_Y(x) \in Y$ is a nearest point to x in Y . If there is no nearest point $y \in Y$ to a given $x \in X$ we use points $y' \in Y$ with $|x - y'| \leq \text{dist}(x, Y) + \epsilon'$ where eventually $\epsilon' \rightarrow 0$. These approximately nearest points y' serve us as well as the actual nearest points. To avoid mess we always use the notation P_Y (rather than $P_Y^{\epsilon'}$ for $\epsilon' \rightarrow 0$). A similar convention will be applied to other functions (such as the length of a curve) which might not assume their minimal values on the spaces in question.

7.3.C Corollary to the Proof of 7.3.A: *The ρ -neighborhood of every geodesic segment in X is 4δ -convex for all $\rho \geq 0$.*

7.3.D Lemma: If X is δ -hyperbolic and $Y \subset X$ is ϵ -convex then

$$|P_Y(x_1) - P_Y(x_2)| \leq \max(2\epsilon + 6\delta, |x_1 - x_2| - |x_1 - P_Y(x_1)| - |x_2 - P_Y(x_2)|)$$

for all x_1 and x_2 in X .

The proof is similar to that of 6.3.A and is left to the reader.

7.3.E Remark: Let us prove a converse to 7.3.D. Namely, let $Y \subset X$ be a subset which admits a retraction $P: X \rightarrow Y$ (i.e. $P|_Y = \text{Id}$), such that the inequalities

$$|x_1 - x_2| \leq 2\delta \text{ and } \max_{j=1,2} |x_j - P(x_j)| \geq \beta,$$

for given positive numbers α and β , imply that $|P(x_1) - P(x_2)| \leq \alpha$ for all points x_1 and x_2 in X . (The map P in 7.3.D satisfies this condition for all $\alpha \geq 2\epsilon + 6\delta$ and $\beta \geq 2\alpha$.) Then the subset Y is ϵ -convex for $\epsilon = 3\beta + \alpha$.

Proof: To estimate $\text{dist}(x, Y)$ for $x \in [x_1, x_2]$, where $y_j \in Y$, $j = 1, 2$, we take a subsegment $[x_1, x_2] \subset [y_1, y_2]$ containing x , such that $|x_i - P(x_i)| \leq \beta$ for $i = 1, 2$ and $|x' - P(x')| \geq \beta$ for all $x' \in [x_1 + \alpha, x_2 - \alpha]$. Observe that $|x_1 - x_2| \geq 2(\text{dist}(x, Y) - \beta)$. Subdivide $[x_1, x_2]$ into k subintervals of length $\leq 2\alpha$ for the minimal possible k and apply the inequality $|P(x'_1) - P(x'_2)| \leq \alpha$ to the end points of these subintervals. This yields

$$|P(x_1) - P(x_2)| \leq |x_1 - x_2| - \text{dist}(x, Y) + \alpha + \beta$$

and

$$|y_1 - y_2| \leq |y_1 - y_2| - \text{dist}(x, Y) + \alpha + 3\beta$$

by the triangle inequality. Q.E.D.

Notice that the above argument does not use any hyperbolicity. In fact, the existence of certain "contracting" maps $X \rightarrow X$ ensures the hyperbolicity of X .

7.3.F Example: Let a metric space X admits a one-parametric family of maps $P_t: X \rightarrow X$ for $t \in [0,1]$, such that

(a) For every curve $S \subset X$ the area of the surface $S \times [0,1] \rightarrow X$ defined by $(s,t) \mapsto P_t(s)$ satisfies

$$\text{area}(S \times [0,1]) \leq C \text{ length } S$$

for some constant C (depending on X but not on S in X).

(b) There exists a constant $C' < 1$ such that

$$\text{area } P_1(D) \leq C' \text{ area } (D)$$

for all surfaces D in X .

Then, obviously, every closed curve S of length $L(S)$ and area $A(S) < \infty$ satisfies

$$A(L) \leq (1-C')^{-1} CL(A),$$

which implies (see 6.8) the hyperbolicity of X , provided

$$L(S) < \infty \Rightarrow A(S) < \infty$$

for all closed curves S in X .

Notice that every convex manifold X admits transformations P_t satisfying (a) with $C = 1$. There are transformations moving every $x \in X$ toward $x_0 \in X$ with unit speed. Furthermore, if X has (strictly) negative curvature $\leq -\epsilon < 0$, then these P_t also satisfy (b), which ensures the hyperbolicity in this case.